of these differences are illustrated by the FORTRAN program listed in Table 12-3. This program uses an algorithm called **decimation in frequency**, while the previously described algorithm is called **decimation in time**. In a decimation in frequency algorithm, the bit reversal sorting is done *after* the three nested loops. There are also FFT routines that completely eliminate the bit reversal sorting. None of these variations significantly improve the performance of the FFT, and you shouldn't worry about which one you are using.

The *important* differences between FFT algorithms concern how data are passed to and from the subroutines. In the BASIC program, data enter and leave the subroutine in the arrays REX[ ] and IMX[ ], with the samples running from index 0 to $N-1$. In the FORTRAN program, data are passed in the complex array $X(\ )$, with the samples running from 1 to N. Since this is an array of complex variables, each sample in X( ) consists of two numbers, a real part and an imaginary part. The length of the DFT must also be passed to these subroutines. In the BASIC program, the variable N% is used for this purpose. In comparison, the FORTRAN program uses the variable M, which is defined to equal $Log_2 N$. For instance, M will be

```
      SUBROUTINE FFT(X,M)
      COMPLEX X(4096),U,S,T
      PI=3.14159265
      N=2**M
      DO 20 L=1,M
      LE=2**(M+1-L)
      LE2=LE/2
      U=(1.0,0.0)
      S=CMPLX(COS(PI/FLOAT(LE2)),-SIN(PI/FLOAT(LE2)))
      DO 20 J=1,LE2
      DO 10 I=J,N,LE
      IP=I+LE2
      T=X(I)+X(IP)
      X(IP)=(X(I)-X(IP))*U
10    X(I)=T
20    U=U*S
      ND2=N/2
      NM1=N-1
      J=1
      DO 50 I=1,NM1
      IF(I.GE.J) GO TO 30
      T=X(J)
      X(J)=X(I)
      X(I)=T
30    K=ND2
40    IF(K.GE.J) GO TO 50
      J=J-K
      K=K/2
      GO TO 40
50    J=J+K
      RETURN
      END
```

TABLE 12-3
The Fast Fourier Transform in FORTRAN. Data are passed to this subroutine in the variables $X(\ )$ and M. The integer, M, is the base two logarithm of the length of the DFT, i.e., $M = 8$ for a 256 point DFT, $M = 12$ for a 4096 point DFT, etc. The complex array, $X(\ )$, holds the time domain data upon entering the DFT. Upon return from this subroutine, $X(\ )$ is overwritten with the frequency domain data. Take note: this subroutine requires that the input and output signals run from $X(1)$ through $X(N)$, rather than the customary $X(0)$ through $X(N\text{-}1)$.